

DOMMARAJU RISHITH VARMA

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Bangalore, Karnataka (Native: Gudur, Nellore District, Andhra Pradesh-524101)

OBJECTIVE

Motivated VLSI engineer seeking to excel in the professional field by applying strong skills in VLSI Physical Design, Synthesis, Floor planning, Power Planning, Placement, CTS, Routing, and STA. Committed to delivering high-quality results through self-motivation, hard work, and effective execution of position requirements.

SKILLS & STRENGTHS

Tool Expertise : DC, ICC2, Genus,

Technical knowledge: Synthesis, PNR, and STA

Scripting languages : TCL, Python

Operating systems : Windows, Linux.

EXPERIENCE

VLSI PHYSICAL DESIGN (Semicon Technolabs Pvt. Ltd., Bengaluru - May 2025 – September 2025)

- Completed a structured training program in VLSI Physical Design covering RTL-to-GDSII flow, design constraints, and implementation methodologies.
- Gained hands-on experience in physical verification, floor planning, placement, routing, and timing closure.
- Worked with scripting languages to automate design tasks and assist in implementation workflows.
- Participated in practical sessions and applied VLSI concepts to real-world design scenarios.
- Demonstrated strong commitment to professional development throughout the training period.

RESEARCH EXPERIENCE PROJECTS

Block-Level Implementation Project (800MHz, 32nm) (Jul 2025 – Sep 2025)

- Utilized ICC2 tool for end-to-end block-level design implementation with a frequency of 800MHz on 32nm technology, handling 40 macros and 159k instances.
- Executed floor and power planning, placement, and clock tree synthesis (CTS); conducted routing and resolved DRC and timing issues.
- Applied advanced PD techniques, iterating multiple floorplan versions to reduce congestion and timing violations.
- Mitigated congestion via strategic placement, cell padding, and blockage management in high-density areas.
- Enhanced timing with path group and bounding adjustments at placement stage; performed manual ECOs for closure of timing and DRV violations.
- Identified and rectified DRC issues during post-route verification

Block-Level Digital Design Implementation Project (500MHz, 28nm) (May 2025 – Jun 2025)

- Led block-level synthesis and physical design of a 500MHz design (28nm technology) using ICC2, DC, Innovus, and Genus tools, with 6 macros and 69,000 instances.
- Managed logic synthesis, floor and power planning, placement, clock tree synthesis, and full routing from floorplan to post-route stage.
- Optimized congestion and achieved target QORs by iterative PD improvements and TCL automation.
- Implemented block-level power planning, resolved PG/DRC and connectivity issues, and handled setup/hold violations analysis and fix

ACADEMIC PROJECT

Green Leaf Disease Detection System (Jan 2025 – Apr 2025)

- Developed an intelligent real-time plant leaf disease detection system leveraging DenseNet121 convolutional neural network (CNN) and computer vision techniques.
- Implemented on Raspberry Pi with edge computing to enable fast, on-device inference for early detection of diseases such as Bacterial Spot, Early Blight, Leaf Mold, and Healthy Leaves.
- Integrated camera module and optimized image processing pipeline ensured accurate classification with minimal latency for real-time operation.
- Designed a portable, low-cost solution catering specifically to remote and small-scale farmers for proactive crop management.
- Features include automatic disease alerts, treatment recommendations, and optional automated spraying mechanism for timely intervention.
- Architected system for future scalability with IoT integration capabilities to enable remote monitoring and data-driven agriculture.

EDUCATION

BTech in Electronics and Communication Engineering
Narayana Engineering College, Gudur, Andhra Pradesh

Dec 2021– Apr 2025
Overall GPA:8.0/10.0

Intermediate
Narayana Junior College, Nellore Dist., Andhra Pradesh

Jun 2019– Mar 2021
Overall GPA: 9.1/10.0

CERTIFICATIONS

- IOT with Micro python issued by Technotran.
 - Digital Image Processing issued by Govt of India (NPTEL)
 - Introduction to Internet of Things issued by Govt of India (NPTEL)
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